Propositional Logic

Properties and Proof in Propositional Calculus

Based on O'Regan's Concise Guide to Formal Methods

Chapter 6 · Propositional Logic

Tautologies **Logical Equivalence** **Laws of Logic** **Substitution** **Proof Calculus** **Natural Deduction**

Tautologies and Contradictions

✓ TAUTOLOGY

Definition: A formula that is TRUE under every possible truth assignment.

Key Insight:
Truth-table shows all T in result column.

Examples:
$P \vee \neg P$ (Law of Excluded Middle)

$P \rightarrow P$

$(P \wedge Q) \rightarrow P$

$\neg P \rightarrow (P \rightarrow Q)$

Notation:
$\models \phi$ means ϕ is a tautology

✗ CONTRADICTION

Definition: A formula that is FALSE under every possible truth assignment.

Key Insight:
Truth-table shows all F in result column.

Examples:
$P \wedge \neg P$ (Law of Non-Contradiction)

$P \leftrightarrow \neg P$

$(P \rightarrow Q) \wedge P \wedge \neg Q$

Notation:
ϕ is unsatisfiable (no model exists)

A formula that is neither a tautology nor a contradiction is called a contingency — its truth depends on the values assigned.

Truth Table Verification

Tautology: P ∨ ¬P

P	¬P	P ∨ ¬P
T	F	**T**
F	T	**T**

Contradiction: P ∧ ¬P

P	¬P	P ∧ ¬P
T	F	**F**
F	T	**F**

Contingency: P → Q

P	Q	P → Q	Meaning
T	T	**T**	Promise kept
T	F	**F**	Promise broken
F	T	**T**	Vacuously true
F	F	**T**	Vacuously true

Rule: If ANY row is F → not a tautology. If ALL rows are F → contradiction. Otherwise → contingency.

Logical Equivalence

Definition: Two formulas P and Q are logically equivalent ($P \equiv Q$) if and only if $P \leftrightarrow Q$ is a tautology — that is, they have identical truth values under every possible truth assignment.

Double Negation

$$\neg\neg P$$

$$\equiv$$

$$P$$

Negating twice restores original

Contrapositive

$$P \rightarrow Q$$

$$\equiv$$

$$\neg Q \rightarrow \neg P$$

Direction flips, values flip

Implication Rewrite

$$P \rightarrow Q$$

$$\equiv$$

$$\neg P \vee Q$$

Implication as disjunction

Verification: Build truth table for $P \leftrightarrow Q$. If every row is T, the formulas are equivalent.

Laws of Propositional Logic (Part 1)

Idempotent Laws

$P \wedge P \equiv P$

$P \vee P \equiv P$

Identity Laws

$P \wedge T \equiv P$

$P \vee F \equiv P$

Domination Laws

$P \vee T \equiv T$

$P \wedge F \equiv F$

Commutative Laws

$P \wedge Q \equiv Q \wedge P$

$P \vee Q \equiv Q \vee P$

Associative Laws

$(P \wedge Q) \wedge R \equiv P \wedge (Q \wedge R)$

$(P \vee Q) \vee R \equiv P \vee (Q \vee R)$

Distributive Laws

$P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$

$P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$

These laws are the algebra of logic — they let us simplify and rewrite propositions without changing meaning.

De Morgan's Laws & Complement Laws

De Morgan's Laws

$$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$$

"NOT (P AND Q)" = "(NOT P) OR (NOT Q)"

$$\neg(P \vee Q) \equiv \neg P \wedge \neg Q$$

"NOT (P OR Q)" = "(NOT P) AND (NOT Q)"

Real-World Example

"It is NOT the case that Alice is at home AND Bob is at home"
≡ "Alice is NOT at home, OR Bob is NOT at home"

Complement: $P \vee \neg P \equiv T$ and $P \wedge \neg P \equiv F$

Double Negation: $\neg\neg P \equiv P$

Substitution in Propositional Logic

Substitution Rule

If $\phi \equiv \psi$ (logically equivalent), then in any formula we may replace every occurrence of ϕ with ψ (or vice versa) and the result is still logically equivalent to the original.

Worked Example: Simplify $\neg(P \wedge Q) \rightarrow R$

Step 1	$\neg(P \wedge Q) \rightarrow R$	*Original formula*
Step 2	$(\neg P \vee \neg Q) \rightarrow R$	*Apply De Morgan: $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$*
Step 3	$\neg(\neg P \vee \neg Q) \vee R$	*Apply: $A \rightarrow B \equiv \neg A \vee B$*
Step 4	$(\neg\neg P \wedge \neg\neg Q) \vee R$	*Apply De Morgan again*
Step 5	$(P \wedge Q) \vee R$	*Apply double negation: $\neg\neg P \equiv P$*

Substitution is the engine of proof: it lets us rewrite formulas step-by-step until we reach a desired form.

Proof in Propositional Calculus: Inference Rules

An inference rule lets us derive new truths from existing ones — the building blocks of formal proof.

Modus Ponens (→E)

$P \rightarrow Q$
P
$\therefore Q$

If it rains, the ground gets wet.
It is raining.
∴ The ground is wet.

Modus Tollens (MT)

$P \rightarrow Q$
$\neg Q$
$\therefore \neg P$

If it rains, ground gets wet.
Ground is NOT wet.
∴ It is NOT raining.

Hypothetical Syllogism

$P \rightarrow Q$
$Q \rightarrow R$
$\therefore P \rightarrow R$

If rain → wet ground.
If wet ground → slippery.
∴ Rain → slippery.

Disjunctive Syllogism

$P \vee Q$
$\neg P$
$\therefore Q$

Pizza or salad.
No pizza available.
∴ Must have salad.

All inference rules are truth-preserving: if premises are true, the conclusion must be true. This is the guarantee of deductive proof.

Formal Proof: A Step-by-Step Example

Goal: Prove that Q follows from premises: P → Q, P ∧ R

Line	Formula	Justification
1	P → Q	Premise (given)
2	P ∧ R	Premise (given)
3	P	Conjunction Elimination (∧E) from line 2
4	Q	Modus Ponens (→E): lines 1 and 3

Reading the proof:

1. Lines 1 & 2 are given premises — no justification needed beyond 'Premise'.
2. Line 3 uses ∧E (And-Elimination): from P ∧ R, we can extract just P.
3. Line 4 is the goal: applying Modus Ponens to P → Q and P gives us Q.

Natural Deduction

Natural deduction (Gentzen, 1935) is a proof system that mimics human reasoning. Each logical connective has INTRODUCTION rules (to build it) and ELIMINATION rules (to use it). Proofs are written as derivation trees.

AND (∧)

Intro	Elim
From A and B, derive A ∧ B (∧I)	From A ∧ B, derive A or B (∧E)

OR (∨)

Intro	Elim
From A, derive A ∨ B (or from B) (∨I)	Case split: if A∨B, and both lead to C → C (∨E)

IF (→)

Intro	Elim
Assume A; prove B; then A → B (→I)	From A→B and A, derive B (→E = MP)

NEGATION (¬)

Intro (¬I): Assume A; derive contradiction (⊥); conclude ¬A. Elim (¬E): From A and ¬A, derive ⊥ (contradiction). Classical: From ¬¬A, derive A.

Curry-Howard Correspondence: Natural deduction proofs correspond to programs in typed lambda calculus. A proposition is a type; its proof is a program of that type. This connects logic and computation at their foundations.

Sequent Calculus (Gentzen's LK System)

Sequent: $\Gamma \vdash \Delta$

A sequent expresses that from the conjunction of formulas Γ (antecedent/premises) one can prove at least one formula from Δ (succedent/conclusions).

Axiom

$A \vdash A$

Every formula proves itself

Cut Rule

$$\frac{\Gamma \vdash A, \Delta \quad \Gamma, A \vdash \Delta}{\Gamma \vdash \Delta}$$

Connects subproofs (like MP)

Left ∧

$$\frac{\Gamma, A, B \vdash \Delta}{\Gamma, A \wedge B \vdash \Delta}$$

Decompose conjunction on left

Right ∧

$$\frac{\Gamma \vdash A, \Delta \quad \Gamma \vdash B, \Delta}{\Gamma \vdash A \wedge B, \Delta}$$

Build conjunction on right

Cut Elimination Theorem (Gentzen): Any proof using Cut can be transformed into a Cut-free proof — fundamental to consistency and decidability.

Soundness and Completeness of Propositional Calculus

SOUNDNESS

Statement:
If $\vdash \phi$ (ϕ is provable),
then $\models \phi$ (ϕ is a tautology).

Meaning:
We cannot prove anything false. Every proof corresponds to a genuine truth.

Slogan:
"Proofs never lie."

COMPLETENESS

Statement:
If $\models \phi$ (ϕ is a tautology),
then $\vdash \phi$ (ϕ is provable).

Meaning:
We can prove everything that is actually true — no truth is out of reach.

Slogan:
"Nothing true goes unprovable."

Key Result (Post, 1921 / Bernays, 1918):

$\vdash \phi \iff \models \phi$ **(Provability ≡ Truth in Propositional Calculus)**

Note: Propositional logic is decidable (SAT is NP-complete). First-order logic is complete but undecidable (Gödel/Church).

Comparing Proof Systems

Feature	Axiomatic (Hilbert)	Natural Deduction	Sequent Calculus
Style	Few rules, many axioms	Many rules, few/no axioms	Symmetric, structural rules
Key Rule	Modus Ponens	Intro/Elim per connective	Axiom + structural rules
Human-readable?	Low	High	Medium
Proof shape	Linear sequence	Tree / derivation tree	Tree of sequents
Main use	Foundations, metalogic	Interactive provers (Lean, Coq)	Theory, cut-elimination
Originators	Frege, Hilbert	Gentzen, Prawitz	Gentzen (LK, 1934)

All three systems are equivalent in expressive power — they prove exactly the same set of tautologies.

Summary

Propositional Logic: Properties and Proof in Calculus

Tautologies & Contradictions

Always true / always false formulas; basis of valid argument forms.

Logical Equivalence

$\phi \equiv \psi$ iff $\phi \leftrightarrow \psi$ is a tautology; used to simplify and substitute.

Laws of Propositional Logic

Idempotent, identity, domination, commutative, associative, distributive, De Morgan, complement.

Substitution

Replacing equivalent sub-formulas — the mechanism of symbolic proof.

Proof in Propositional Calculus

Modus Ponens, Modus Tollens, Hypothetical & Disjunctive Syllogism.

Natural Deduction

Intro/Elim rules per connective; proofs as trees; Curry-Howard correspondence.

Sequent Calculus

$\Gamma \vdash \Delta$ judgments; Cut rule; Cut Elimination Theorem (Gentzen).

Soundness & Completeness

$\vdash \phi \Leftrightarrow \models \phi$ — propositional calculus proves exactly the tautologies.